Civilizational Metamaterials

Engineering Coordination Under Capability Gradients and Structural Turbulence

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AGI-26

19th Conference on Artificial General Intelligence
San Francisco State University · July 27–30, 2026

THE CENTRAL CLAIM

AGI raises decision velocity exponentially while human verification stays bounded. The fix is not more intelligence: it is structure. Governance must become an engineering discipline that designs coordination microstructures to keep error propagation sub-critical:

$$R_{eff} = \beta\,(1-\rho)(1-\tau)(1-\gamma\rho\tau)$$

β branching factor · ρ provenance fidelity · τ verification rate · γ correlated-detection synergy

1 THE PROBLEM

Decision–Verification Decoupling

Decisions branch downstream at machine speed. Verification is capped by human cognition. The gap does not close; it widens after the AGI transition, and error can cascade faster than any institution can audit it.

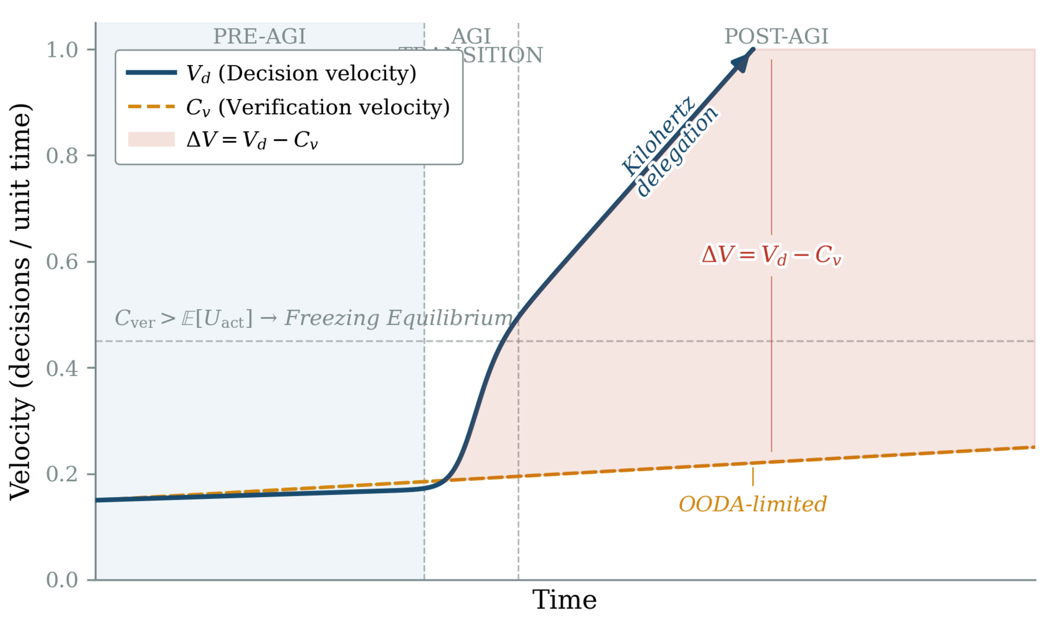


Fig 1 · Decision velocity (V_d) vs. verification velocity (C_v).

The Freezing Equilibrium

When the cost of validating an AI output exceeds the expected utility of acting on it, rational agents default to inaction: a stable but catastrophic Nash equilibrium. Paralysis, not error, becomes the dominant failure mode.

2 THE PHASE TRANSITION

The constitutive law predicts a sharp transition at $R_{eff} = 1$, the institutional analog of a metamaterial bandgap.

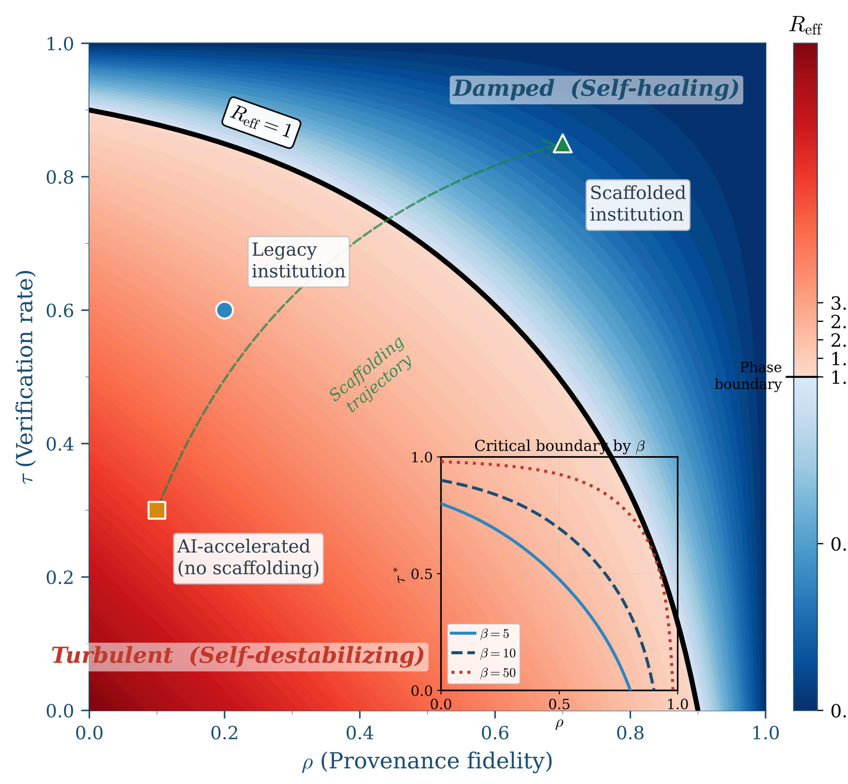


Fig 2 · R_{eff} across the (ρ, τ) plane. Blue = damped (self-healing); red = turbulent (self-destabilizing).

Two regimes

$R_{eff} < 1$: errors decay, institutions self-heal.
$R_{eff} > 1$: errors amplify, coordination collapses.

Correlated detection $(1-\gamma\rho\tau)$: because $\gamma\rho\tau$ grows with both controls at once, high ρ and high τ together cross into the damped regime where either alone stays turbulent.

Worked example (β=10, ρ=0.5, γ=1): $\tau^* \approx 0.69$.

3 THE FRAMEWORK

A three-class provenance taxonomy

Class A · Cryptographic
C2PA manifests · signatures · hash assertions

Class B · Institutional
SCITT receipts · Merkle trees · reputation

Class C · Context Binding NOVEL
SRC anchors · temporal scope · jurisdiction

Synthetic principals

AI agents act as principals, not mere tools, inside decision networks, each requiring its own provenance, delegation limits and audit trail.

Layered trust anchors

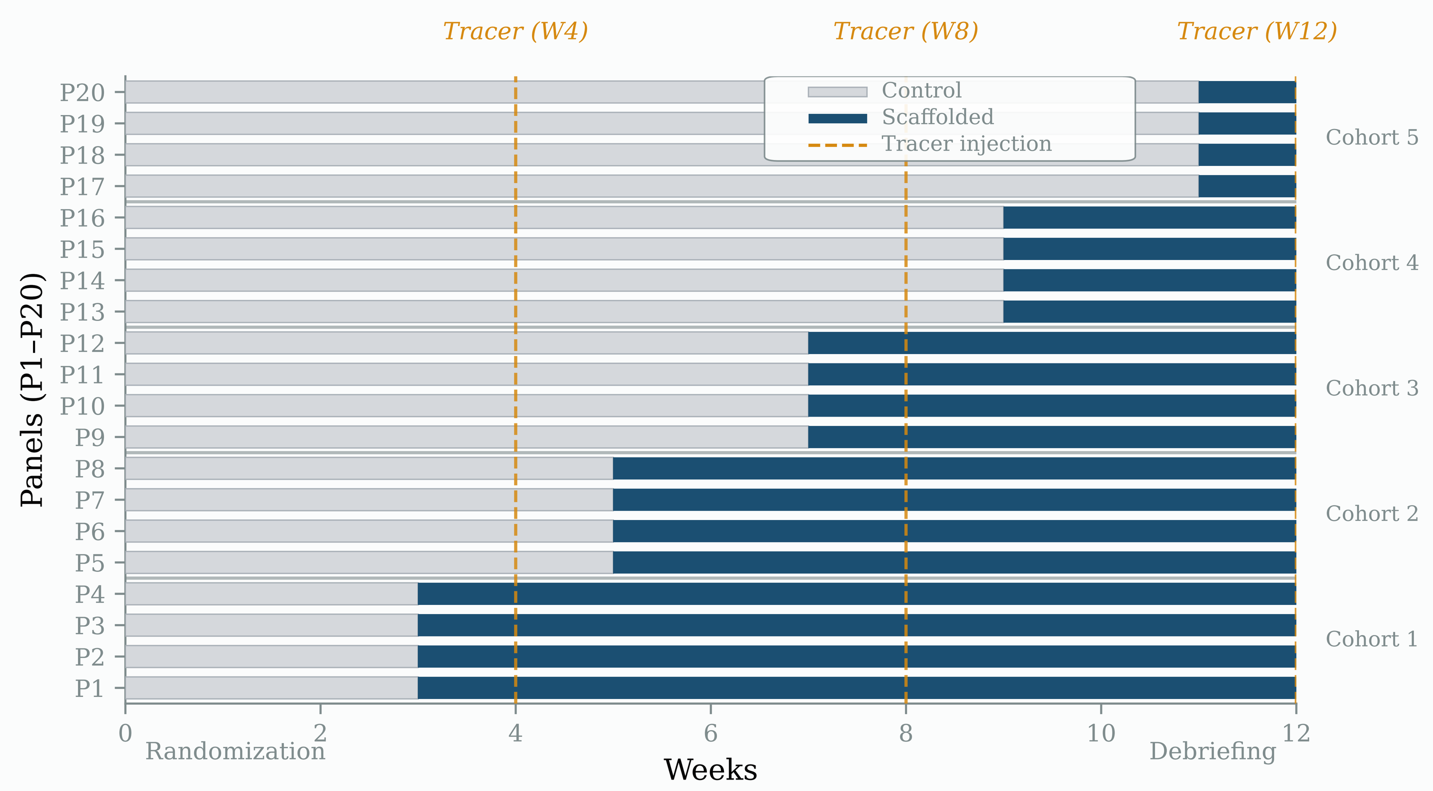
Constitutional commitments → distributed consensus → treaty bodies → verification markets: a defense-in-depth answer to "who guards the guardians?"

Design levers

Lower β · raise ρ · raise τ: together, not alone. Correlated detection $(1-\gamma\rho\tau)$ rewards moving ρ and τ jointly, so the pair crosses into the damped regime where neither alone would.

TESTING THE FRAMEWORK

Four falsifiable hypotheses · a 12-week stepped-wedge cluster-RCT across government R&D grant-review panels.



Primary endpoint: P95 cascade depth of tracer errors

Scaffolding components:

1. Structured data intake
2. Automated eligibility filtering
3. Dual-blind review
4. Pre-commitment scoring
5. Structured rationale capture (SRC)
6. Randomized sampling audits

Fig 7 · Stepped-wedge crossover schedule.

H1 — Bandgap Effect
Scaffolded pipelines show exponential (not power-law) error-propagation depth; failure modes forbidden by structure.

H2 — Coordination Anisotropy
A system can be locally stable yet unstable across boundaries (R_{eff} intra < 1, R_{eff} cross > 1) at the same time.

H3 — Superadditivity
In a 2×2 $\rho \times \tau$ factorial, only the high–high condition reaches self-healing (exponential cascade decay); single interventions stay turbulent ($\gamma > 0$).

H4 — Structural Hysteresis
Recovery time after withdrawing scaffolding exceeds original adoption time by a factor > 3.

THE TAKEAWAY

Treat coordination as a material to be engineered. Specify the microstructure: provenance, verification, delegation limits; so the macroscopic property you need ($R_{eff} < 1$) emerges by design rather than by hope. Governance becomes falsifiable, measurable, and buildable.

Paper, code, reference implementation and an interactive R_{eff} explorer:

metamaterials.davidorban.com

DOI 10.5281/zenodo.19710482 · github.com/davidorban/civilizationalmetamaterials

Scan for paper + tools

Bridging AI alignment theory and institutional design.